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A PROMPT-BASED METHOD WITH MULTI-VIEW OPTIMIZATION FOR OPEN RELATION EXTRACTION

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ABSTRACT

Open Relation Extraction (OpenRE) is a task that involves discovering new relation types by referring to labeled instances. Existing methods mainly rely on large pre-trained models to obtain the relation representation of entity pairs, and then jointly train the supervised and unsupervised data using a joint loss function. Some researchers enhance their relation representation by introducing additional information as prompt. However, these approaches have several major issues. Firstly, many of them rely on external knowledge bases, which require a large amount of high-quality data. Secondly, They fail to consider the rich semantic and prior knowledge existing in the labels. To this end, we propose a novel method to incorporate the prior knowledge into prompt-tuning and introduce a multi-view algorithm to optimize the relation representations. We conduct experiments on two common datasets, and the results show that our proposed method significantly outperforms the previous state-of-the-art methods.

***Index Terms*—** Relation Extraction, Prompt, Knowledge Injection, Open Domain

1. INTRODUCTION

Relation extraction (RE) is the task of identifying the type of relation between a pair of entities, which is crucial for the automatic construction of knowledge bases. Traditional RE methods [1–3] mostly rely on a significant amount of supervised data to learn the distribution of text. However, the absence of high-quality supervised data poses a challenge, particularly in professional domains such as medical research. In recent years, many researchers focus on the task of open-domain relation extraction (OpenRE), using a portion of labeled data to teach the model to learn universal expressions of relations, allowing it to discover new types in new corpora.

[4] first propose Relational Siamese Networks (RSN) to tackle the challenge of relation extraction. It learns similarity metrics from labeled data and transfer relational knowledge to identify novel relations in unlabeled data. [5] utilize weak self-supervised signals and large pre-trained language models to perform adaptive clustering and enhance contextualized features for relation classification. Recent works make significant progress in obtaining high-quality pseudo-labels using various mechanisms. However, few of them explore the idea of prompt-tuning and fully utilize the rich semantic information contained in pre-defined labels. Although some researchers apply the idea of prompt-tuning to address the OpenRE task such as [6], they fail to fully leverage the various types of information conveying by the labels for the semantic space. [7] utilize entity information stored in external knowledge base to enhance relation representation, however, this approach requires additional relevant data support, resulting in low resource utilization.

Due to the limited annotated data in OpenRE, it is important to explore existing language patterns to discover novel relations in new corpora. Inspired by [8], in this paper we propose a novel method to extend the potential semantic space by leveraging the generative ability of PLMs such as BERT [9]. By parsing the labels in datasets, we obtain useful auxiliary information to improve the expression of relations in sentences. For instance, in the Fewrel dataset [10], the relation label "head of government" refers to the highest-ranking official in a government who is primarily responsible, and the implied entity types of subject and object can be identified with high confidence as "person" and "government institutions". However, there is typically no single word that is an exact equivalent of "head of government", terms such as "leader" or "chief" can refer to a person in a leadership position within an organization or country, but this also deviates from the original meaning. We encode the original sentence and the concatenated prompt, then we construct vector representations in the vocabulary that have the same dimensions as the tokens which fully express the relation information. Finally, we optimize the relation representation from both the perspectives of the entity pairs and the `[mask]` token in the prompt. Experimental results demonstrate that our proposed improvement achieves state-of-the-art performance.

2. METHOD

We separate labels into two parts: a labeled dataset $\mathcal{D}^l$ with pre-defined relation types and an unlabeled dataset $\mathcal{D}^u$ with unseen relation types. Note that the types for these two sets

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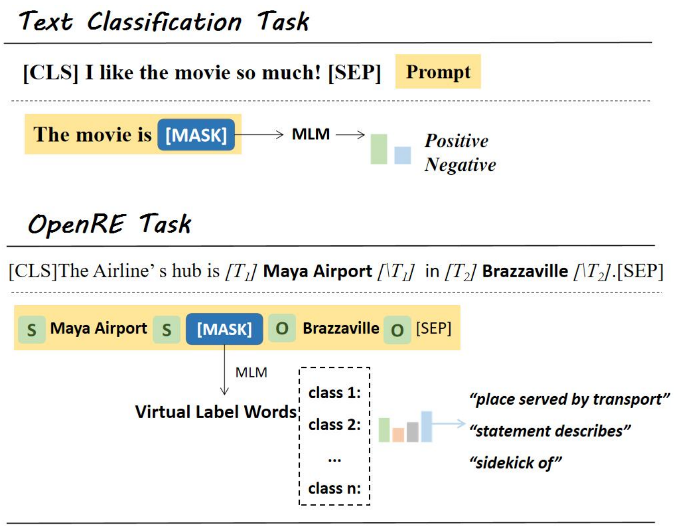

Fig. 1 Examples of prompt-tuning in different tasks.

are mutually exclusive.

2.1. Knowledge Injection And Sentence Encoder

Inspired by [8], in this work we extract useful information from the labels. For example, the label "head of government" can provide two valid entity types: "person" and "government". We insert type markers around the entities and initialize the markers with prior knowledge implied in the predefined labels. Additionally, we reconstruct the semantic space of PLMs by adding virtual words that can fully express the meaning of relation labels. We wrap the sentence by concatenating the template $\mathcal{T}$, encouraging the model to generate appropriate representations at `[MASK]`, `[sub]`, and `[obj]`. $\mathcal{T}(\cdot)=$ "`[sub]` $\mathbf{h_i}$ `[sub]` `[MASK]` `[sub]` $\mathbf{t_i}$ `[sub]`", the input can be represented as: $X_{prompt}=$ `[CLS]` X_{in} `[SEP]` $\oplus\ \mathcal{T}$.

where X_{in} represents the original tokens in the sentence. We feed X_{prompt} into PLM to obtain the initial representation of the head and tail entities, as well as the `[MASK]` tokens. We concate the head and tail entity embeddings to obtain the representation h_i as token-view representation and take the probability distribution at `[MASK]` $p(y^i_{msk}|x)$ as mask-view representation.

$$p(y^i_{mask}|x)=P(\texttt{[MASK]}=v|x_{prompt}) \tag{1}$$

where i means i^{th} instance in training.

2.2. Multi-View Optimization Module

To train both token-view and mask-view features, we use a non-linear network to obtain low-dimensional representations $h^{l'}$, which accurately reflect the relational semantic similarity. The output features from the PLM are fed into this network, and we optimize model by minimizing the distance between the cluster centroid c and the token-view relation representations.

$$\mathcal{L}_c=\frac{1}{2N}\sum_{i=1}^{N}\|h_i^{l'}-c_{y_i}\|_2^2 \tag{2}$$

where c_{y_i} means the cluster centroid.

For `[MASK]` view, we take the probability distribution at the `[MASK]` position and use cross-entropy to jointly optimize the `[sub]`, `[obj]`, and `[MASK]` embeddings, aligning them with the meaning of entity types and relations.

$$\mathcal{L}_{\texttt{[MASK]}}=-\frac{1}{N}\sum_{i=1}^{N}y_i\log p(y^i_{mask}|x_i) \tag{3}$$

where y_i denotes the groud-truth of instance x_i.

To optimize the relation representation at `[MASK]` token, we design an additional constraint loss by leveraging the implicit constraint between the entity pairs and the relation between them. Specifically, given a (e_h, r, e_t) to represent a sample, where e_h and e_t represent the head and tail entities, respectively, we compute the average of all tokens involved in e_h and e_t, resulting in an entity pair representation (e_h, e_t), with the relation between them being denoted by r which derived from the `[MASK]` token. We define the following loss to use the implicit constraints and enhance the interaction between two views:

$$\begin{aligned}\mathcal{L}_{constraint}=&-\log\sigma(\gamma-d_r(e_h,e_t))-\\&\sum_{i=0}^{n}\frac{1}{n}\log\sigma(d_r(e_h^{i'},e_t^{i'})-\gamma)\end{aligned} \tag{4}$$

$$d_r(e_h,e_t)=\|e_h+r-e_t\|_2 \tag{5}$$

We randomly replace the e_h and e_t in the triplet with unrelated type representations$(e_h^{i'}, e_t^{i'})$. and enhance the correlation between entity representations and relations by $\mathcal{L}_{constraint}$. Where γ is the margin, σ refers to the sigmoid function and d_r is the scoring function.

Finally, we apply the model to unlabeled instances to obtain the relation representation $h^{u'}$, and use k-means algorithm to cluster the unlabeled data to obtain pseudo-labels for samples.

$$y^u=k-means(h^{u'}) \tag{6}$$

where y^u means the pseudo-labels generated by clustering.

2.3. Relation Classification Module

After obtaining the pseudo-labels, we evaluate the quality of the generated labels by KL-divergence and optimize the representations by reducing the intra-cluster distance and increasing the inter-cluster distance. We use $\overrightarrow{s_i^u}$ to represent the probability distribution of sentence s_i^u and $\overrightarrow{s_j^u}$ to represent the

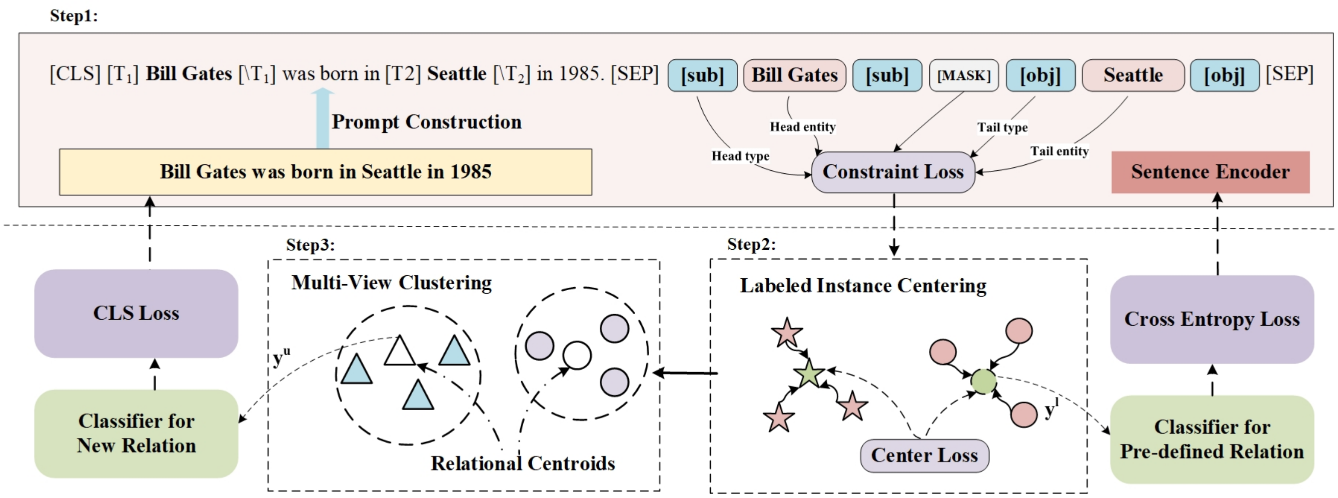

Fig. 2 Overview of our proposed model. We present an example to illustrate the process of constructing a prompt and feeding it into the encoder. The pink label represents the entity pair involved, and the blue label indicates the type of the entity. We optimize the loss generated by both the mask token and the entity token, and apply the model in unlabeled data to transfer the learned knowledge.

probability distribution of sentence s_j^u. The KL-divergence between them is defined as:

$$\mathcal{L}^+(s_i^u, s_j^u) = KL(\overrightarrow{s_i^u} \| \overrightarrow{s_j^u}) + KL(\overrightarrow{s_j^u} \| \overrightarrow{s_i^u}) \tag{7}$$

We use hinge loss to train the model when they come from different clusters:

$$\begin{aligned} \mathcal{L}^-(s_i^u, s_j^u) = L_h(KL(\overrightarrow{s_i^u} \| \overrightarrow{s_j^u}), \sigma) + \\ L_h(KL(\overrightarrow{s_j^u} \| \overrightarrow{s_i^u}), \sigma) \end{aligned} \tag{8}$$

$$L_h(e, \sigma) = max(0, \sigma - e) \tag{9}$$

we set a symbol d_{ij} to denote whether s_i^u and s_j^u come from the same cluster.

$$b_{ij} = \begin{cases} 1, & \text{if } p_i = p_j \\ 0, & \text{if } p_i != p_j \end{cases} \tag{10}$$

where p_i, p_j denote the pseudo-label of s_i^u and s_j^u. The final loss can be representened as:

$$\begin{aligned} L_{CLU}(s_i^u, s_j^u) = b_{ij}\mathcal{L}^+(s_i^u, s_j^u) \\ +(1 - b_{ij})\mathcal{L}^-(s_i^u, s_j^u) \end{aligned} \tag{11}$$

2.4. Training Procedure

Based on the supervision signal provided by the pre-defined labels, we train a classifier $\mathcal{C} : R^d \rightarrow R^c$ to classify and optimize the relation representation h^l. We use cross-entropy to optimize $\mathcal{C}$, which is defined as:

$$\mathcal{L}_{CE} = -\frac{1}{N} \sum_{i=1}^{N} \log \mathcal{C}_{y_i}^l(h_i) \tag{12}$$

We continuously optimize the performance through iterative training to improve the quality of pseudo-label generation. The overall loss as follows:

$$\mathcal{L}_{CLS} = \mathcal{L}_{CE} + \mathcal{L}_{CLU} + \mathcal{L}_{\texttt{[MASK]}} \tag{13}$$

3. EXPERIMENTS

3.1. Datasets

We evaluated our method on two commonly used public datasets, FewRel and TACRED [10, 11]. We followed the setting in [12] and used the training set as the labeled set, and the validation set as the unlabeled set. For testing, we randomly selected 1,600 instances from the unlabeled set. TACRED contains 41 relation types. We selected 31 relations as the labeled set and 9 relations as the unlabeled set. For testing, we randomly selected 15% of the instances from the unlabeled set.

3.2. Compared Models

For fair comparison, we select six prevailing OpenRE models as follow:

Etype+ predict the relation types based on the concatenation of two entity types. **SelfORE** utilize self-supervised signals through PLMs for adaptive clustering of contextual features. **RSN** first propose to transfer the knowledge from annotated data to new corpora. **RSN-BERT** a modified version of RSN, where the static word vector was substituted with

Dataset	Method	B^3	V-measure	ARI
FewRel	RSN [4]	0.589	0.708	0.453
	EType+ [13]	0.319	0.408	0.249
	SelfORE [5]	0.678	0.783	0.647
	RoCORE [12]	0.796	0.860	0.709
	TABS [6]	0.761	0.837	0.727
	Ours	**0.807**	**0.868**	**0.740**
TACRED	RSN [4]	0.631	0.643	0.459
	EType+ [13]	0.439	0.364	0.143
	SelfORE [5]	0.541	0.619	0.447
	RoCORE [12]	0.860	0.888	0.812
	TABS [6]	0.854	0.872	**0.828**
	Ours	**0.863**	**0.889**	0.816

Table 1 We conducted three rounds of experiments on the FewRel and TACRED datasets using different random seeds, and calculated the average of the results as the final experimental performance. The experimental results of other baselines are obtained from [12] and [6].

model	Prec.	B^3	V-measure	ARI
Rocore	0.647	0.679	0.747	0.546
Ours	**0.747**	**0.752**	**0.811**	**0.646**

Table 2 The performance on the cross-domain test set of FewRel.

Label	Top 3 words around `[MASK]`
tributary	river, flows, water
father	heirjo, father, himselfrid
developer	research, renault, design

Table 3 The best matched word in vocabulary of `[MASK]`. The words are sorted by frequency of appearance in top 3 predictions. We skip the virtual label words(`[class i]`).

BERT embeddings. **RoCORE** propose to align the generated clusters with relational semantics, and reduces bias through joint iterative training. **TABS** propose a prompt-based learning framework.

3.3. Implementation Details

We use BERT-base as the encoder and follow the approach outlined in [12] by using the 8th layer and freezing the other layers to avoid overfitting. Our experiments are conducted on a GeForce GTX 3090 with 24GB of memory.

3.4. Main Results

Based on the results reported in Table 1, our proposed method achieves the state-of-the-art performance. To verify the generalizability of our method, we conduct experiments on FewRel

model	Prec.	B^3
Full Model	0.761	**0.807**
w/o constraint loss	0.760	0.803
w/o knowledge injection	0.757	0.791
w/o prompt	**0.765**	0.796

Table 4 Ablation study of our method on FewRel. We separately validate the effect of constraint loss, knowledge injection and prompt.

2.0 which provides a test set from the medical domain. The results are presented in Table 2. Based on the experimental results, our method is significantly better than Rocore [12]. In this experimental setting, we only did one epoch of pre-training. We believe the reason for the better performance is that Rocore relies more on the knowledge transfer process during the pre-training phase, while our method does not.

3.5. Ablation Study

We conduct ablation experiments to separately evaluate the effect of the prompt module, knowledge injection module and constraint loss. We show the statics in Table 4. The Full Model achieve a Precision of 0.761 and an F1 of 0.807. Both the removing of constraint loss and knowledge injection module results in a decrease in Precision and F1. It indicates that the knowledge injection module plays a more important role in the model. Based on the results, it shows that the addition of prompt alone yield opposite effect without the knowledge injection at the `[mask]` position, we think that unconstrained prompt may introduce a kind of noise into the model, which affects the performance of the experiment.

3.6. Other Analysis

Moreover, to understand the significance of the `[mask]` token, we obtain the hidden state of `[mask]` and search for the word in the vocabulary that has the closest feature. The results are shown in Table 3. We discover that the embedding of the `[mask]` has semantic consistency with the relation labels, resembling the prototype representation of the relations.

4. CONCLUSION

In this paper, we introduce a novel method for OpenRE and evaluate its performance on two widely used datasets, FewRel and TACRED. Our method improves in two aspects: We consider multi-view representations and optimize both of them simultaneously to improve relation prediction. (2) We inject prior knowledge into prompt and use the implicit constaint to enhance the interaction between two views. The experimental results on two datasets demonstrate the effectiveness of our method.

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